



HD in Computing & Software Engineering

CSE5015 Computing Project Proposal Approval Sheet

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Group Name	Group 1	Attempt No	3
Title of the project	HappyPaws: A Mobile Application for Animal Rescue and Rehoming		
System platform	☐ Web Application ☐ Desktop Application ☒ Mobile Application ☐		
Technology	Kotlin (Android), React/TypeScript (Web), ASP.NET Core/C# (Backend), PostgreSQL, AWS, and Gemini AI		
Programming methodology	☐ Structured System Analysis and Design Methodology ☒ Object Oriented Analysis and Design Methodology		
SDLC	Agile		

Student No	GM/HDCSE/CMU/18/01	Full Name	I P Hashan Dimuthu Bandara
Roles & Responsibilities:			
Project Manager / Scrum Master Handles agile delivery, Jira tracking, requirements analysis, documentation, and coordinates the sprints.			
Student No	GM/HDCSE/CMU/18/04	Full Name	D I T Nethmina Gunasekara
Roles & Responsibilities:			
Lead Developer (Backend, Mobile & DevOps) Handles ASP.NET Core API, PostgreSQL database, Gemini API integration, and AWS deployment.			
Student No	GM/HDCSE/CMU/18/09	Full Name	P Tharushi Hiranya Sandakalpani
Roles & Responsibilities:			
UI/UX Designer & Frontend Developer Creates Figma mockups, designs user flows, and develops the React/TypeScript administration panel.			
Student No	GM/HDCSE/CMU/18/12	Full Name	Mandari Perera
Roles & Responsibilities:			
Quality Assurance (QA) & Documentation Specialist Tests Android app and admin panel, reports bugs, writes system documentation and user guides.			



Student No	GM/HDCSE/CMU/18/08	Full Name	Vishwa Sithija Ransara
Roles & Responsibilities:			
Business Analyst (BA) & Testing Assistant			
Creates UML diagrams, prepares sample data, assists with QA testing, and researches adoption rules.			

Adopter and owner functionalities:

Functionality	Description	Justification
Registration with role selection	Creates an account and selects the adopter or owner role during sign-up, before any listing, messaging, or application function becomes available.	Role selection at registration is what lets the platform show each user only the functions suited to their part in the rescue and rehoming process, which is the starting point of the trust model.
Login and authentication	Signs in securely, with role-based access enforced through JWT so the session only exposes adopter-facing screens and data.	Secure, role-scoped access keeps one account from reaching another role's verification queue or case data, which matters once reputation is tied to identity.
Identity verification (KYC)	Submits a national identity document that an administrator reviews and approves through the web panel before the account can list, apply, or message freely.	A verified adopter is what separates a committed applicant from an anonymous one, which is the specific gap this proposal argues existing tools leave open.
Lifestyle profile and matching	Completes a short profile covering home size, activity level, and existing pets, and receives animal suggestions from a rule-based matching engine built on that profile.	Matching on lifestyle rather than availability alone raises the odds that a placement lasts, and a transparent rule set is easier to explain to an examiner and a user than a trained model would be.
Browse and search adoption listings	Searches and filters listed animals by species, location, and urgency, independent of any suggestion the matching engine has already made.	Not every adopter arrives with a completed lifestyle profile, so open browsing keeps the adoption hub usable from the first visit.
Adoption listing (owner)	An existing owner posts their own animal for rehoming with photos and a description, using the same listing flow an administrator can moderate.	Owners rehoming a pet directly are a distinct case from a rescued animal placed by a foster, and the platform has to support both without treating one as an edge case of the other.
Submit and track an adoption application	Applies to adopt a listed animal and follows the application through pending, accepted, or declined status from the same screen.	Status tracking closes the loop on an application that would otherwise depend on a private message the adopter has no way to confirm was seen.
Rescue reporting	Photographs an injured or stray animal encountered in the field while the app captures the device's GPS location automatically, and submits the report for triage.	Any verified user, not only fosters or transporters, can be the first to find an animal in distress, so the reporting path has to stay open to every role.
Private in-app messaging	Communicates with a foster, owner, or administrator through encrypted in-app chat built on SignalR, without exchanging a phone number.	Removing the phone number from the exchange is the specific protection this proposal identifies as missing from Facebook groups and classifieds.

Reputation and trust badges	Accumulates reputation points and visible trust badges from confirmed activity such as a completed adoption or a verified rescue report.	Reputation is the mechanism that carries an adopter's history from one interaction to the next, which is the argument this proposal makes for why the platform is more than a listing board.
Notifications	Receives push alerts, through Firebase Cloud Messaging, for new messages, application status changes, and nearby rescue cases that need a verified responder.	An adopter who is also a potential nearby responder should not have to keep the app open to learn that a case near them needs help.

Foster functionalities:

Functionality	Description	Justification
Registration with role selection	Creates an account and selects the foster role, which unlocks the case-acceptance and animal-handover functions once verified.	Fostering means taking a rescued animal into a private home, so the platform has to know a user occupies this role before it can route a case to them.
Login and authentication	Signs in securely with role-scoped, JWT-enforced access to foster-facing case and messaging screens.	The session should expose only what a foster needs to act on a case, in line with the same access model every role follows.
Identity verification (KYC)	Submits an identity document for administrator review before the account can accept a placement or receive geo-targeted alerts.	A foster takes physical custody of a rescued animal, so verifying the person behind the account carries more weight here than it does for a role limited to browsing or messaging.
Geo-targeted rescue alerts	Receives a push notification, through Firebase Cloud Messaging, when a nearby triaged case needs a temporary foster placement.	A foster can only help if the alert reaches them while the case is still current, which is why alerting is tied to location rather than a general broadcast.
Rescue reporting	Photographs of an injured or stray animal found while out, with the device recording the location automatically for triage.	A foster is often the person already on the road near an incident, so the reporting capability applies to this role as directly as it does to any other user.
Foster placement acceptance and case handling	Accepts a rescue case referred to them, views the case on the live map and case screen, and updates the animal's condition and foster status as it changes.	The platform needs a record of who currently holds an animal and in what condition, which is what keeps the live case map useful to an administrator coordinating several cases at once.
Transition to adoption listing	Once a fostered animal is ready to be rehomed, creates or hands off an adoption listing so the adoption hub can match it with an adopter.	This is the link between the rescue radar module and the adoption hub, and without it a fostered animal has no path back into the system once it recovers.
Private in-app messaging	Coordinates handover timing and animal condition with transporters, veterinarians, and prospective adopters through encrypted in-app chat, without sharing a phone number.	A foster typically has to coordinate with more roles than any other user in a single case, which makes private messaging a functional necessity here rather than a convenience.
Reputation and trust badges	Builds reputation and a visible trust badge from completed foster placements and verified rescue reports.	A foster's track record is exactly the kind of history an adopter or administrator would want visible before trusting them with an animal, which is the reputation



		argument applied to this role specifically.
Notifications	Receives alerts for new nearby cases, incoming messages, and status changes on an animal currently in their care.	A foster is responsible for an animal in real time, so a missed alert has a more immediate cost for this role than for one limited to browsing listings.

Transporter functionalities:

Functionality	Description	Justification
Registration with role selection	Creates an account and selects the transporter role, which unlocks transport-request alerts and case-status updates once verified.	Transport tasks involve carrying another user's animal, so the role has to be declared and confirmed before the platform routes a request to that account.
Login and authentication	Signs in securely with role-scoped, JWT-enforced access to transport requests and case screens.	Restricts what the session can see and act on to the transport function this role exists to perform.
Identity verification (KYC)	Submits an identity document for administrator review before the account can accept a transport task.	A verified transporter is what lets a foster or veterinarian hand over an animal without relying on an unverified stranger from a group chat, which is precisely the failure mode described in the problem statement.
Geo-targeted transport alerts	Receives a push notification when a nearby case needs an animal moved between a rescue site, a foster, a veterinarian, or an adopter.	Transport only helps if it reaches someone close enough to respond quickly, so this alert is scoped by location in the same way the rescue alert is.
Rescue reporting	Reports an animal found while en route, with photo and automatic location capture feeding the same triage pipeline every other role uses.	A transporter spends more time moving through the areas where strays are found than most other roles, so excluding them from reporting would waste that exposure.
Accept and track a transport task	Claims an open transport request and updates its status through assigned, picked up, in transit, and delivered, which the administration panel's case map reflects in real time.	Everyone downstream of a transport, the foster expecting a drop-off, the vet expecting an arrival, the administrator watching the case, needs the same status without having to ask for it directly.
Private in-app messaging	Coordinates pickup and drop-off timing with fosters, veterinarians, and adopters through encrypted chat, without exchanging a phone number.	Transport logistics depend on precise timing, and in-app messaging keeps that coordination inside the same trust boundary as everything else on the platform.
Reputation and trust badges	Builds reputation and a visible trust badge from completed, verified transport runs.	A transporter's reliability is exactly what a foster or veterinarian needs to see before trusting them with an animal in transit, which reputation makes visible instead of assumed.
Notifications	Receives alerts for open transport requests nearby, new messages, and status updates on any case currently assigned to them.	A transporter working multiple cases needs to know the moment a new request appears near their route, which is what keeps the



		transport layer responsive rather than passive.
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Sponsor and donor functionalities:

Functionality	Description	Justification
Registration with role selection	Creates an account and selects the sponsor or donor role, which unlocks case browsing and sponsorship tracking once verified.	A sponsor's credibility, like every other role's, starts with a declared and verified identity rather than an anonymous pledge.
Login and authentication	Signs in securely with role-scoped, JWT-enforced access to sponsorship and case-tracking screens.	Keeps a sponsor's session limited to the browsing and tracking functions this role needs, consistent with every other role on the platform.
Identity verification (KYC)	Submits an identity document for administrator review before the account can pledge support against a case or listing.	The proposal names fraudulent medical appeals as one of the specific failures a verified identity is meant to close, and a verified sponsor is the other side of that same fix.
Browse rescue cases and adoption listings	Views triaged rescue cases and adoption listings, including the AI urgency classification attached to each one, to decide who or what to support.	A sponsor needs the same case information a responder sees in order to direct support toward the cases that need it most, not just the ones that are most visible.
Pledge support to a case or listing	Records a sponsorship commitment against a specific animal or case. In the three-week MVP this is a simulated pledge rather than a live transaction, consistent with the proposal's decision to defer real payment processing to the post-MVP roadmap.	Demonstrating the sponsorship flow end to end matters for the viva even though the payment itself is deferred, because it shows where the direct-to-vet funding feature will attach once it is built.
Track sponsored case progress	Follows status updates on an animal or case the sponsor has backed, from the same screen used to make the original pledge.	A sponsor who cannot see what happened after a pledge has no way to judge whether the platform's claim of transparency holds up, which is the trust argument applied to donations specifically.
Private in-app messaging	Contacts the foster, veterinarian, or administrator attached to a sponsored case for an update, without sharing a phone number.	Sponsors are as entitled to the platform's privacy protection as any other role, even though their interaction with a case is usually more distant than a foster's or a transporter's.
Reputation and trust badges	Builds reputation and a visible trust badge from confirmed, completed sponsorships.	A visible sponsorship history is what turns a one-off donor into a recognised supporter other users can see and trust, extending the reputation model to the donation side of the platform.



Notifications	Receives alerts for updates on sponsored cases and incoming messages.	A sponsor who has committed support to a specific animal has a direct interest in knowing what happens to it next, which notifications close without requiring the sponsor to check back manually.
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Veterinarian functionalities:

Functionality	Description	Justification
Registration with role selection	Creates an account and selects the veterinarian role, which unlocks medical case review and advisory messaging once verified.	The platform needs to know which accounts belong to a qualified professional before it can route a medical review or display an expert badge.
Login and authentication	Signs in securely with role-scoped, JWT-enforced access to case review and messaging screens.	Keeps a veterinarian's professional-facing access separate from the general user session every other role receives.
Identity verification (KYC)	Submits a clinic registration or professional license, rather than a general identity document, for administrator review before the account can act as a verified veterinarian.	Medical credibility on the platform depends on a checked credential rather than a claimed title, which is what makes a veterinarian's advice worth more than a comment from whoever answers first.
Geo-targeted rescue alerts	Receives a push notification when a nearby case is classified as critical or moderate and needs medical attention.	Routing medical cases to a nearby verified veterinarian is what turns AI triage into an actual response rather than a label with nowhere to go.
Review AI photo triage and confirm urgency	Opens a reported case, reviews Gemini's critical, moderate, or low classification against the submitted photo, and confirms or overrides it with a professional judgement.	The proposal treats the AI's output as advice rather than instruction, and a veterinarian's confirmation is the professional check that keeps a wrong classification from becoming a wrong decision, alongside the same override available to an administrator.
Provide medical guidance	Responds to a case or a private message with treatment advice for the animal's foster, transporter, or owner.	The problem statement names unqualified advice, arriving from whoever comments first, as one of the specific failures a verified veterinarian role is meant to correct.
Private in-app messaging	Communicates directly with the user currently handling an animal, without a shared phone number, to give guidance or ask for more information.	Medical advice often needs a back-and-forth exchange, and keeping that inside encrypted in-app chat protects both the vet's and the user's contact details.
Reputation and trust badges	Builds reputation and an expert-verified badge from confirmed medical input and completed case reviews.	A visible, earned expert badge is what lets a user tell a genuine veterinarian's advice apart from an unverified comment, which is the exact distinction the problem statement says current channels cannot make.



Notifications	Receives alerts for new nearby cases that need medical review and incoming messages.	A veterinarian who is not prompted the moment a critical case appears nearby cannot act on the urgency the AI triage step already identified.
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Administrator functionalities:

Functionality	Description	Justification
Login and authentication	Signs in to the web administration panel with elevated, role-based access enforced through JWT.	Administrator access reaches every other role's verification and case data, so it needs the strictest access control on the platform.
Identity verification review	Inspects a submitted identity document, clinic registration, or license for any role and approves or rejects it before the account gains full privileges.	Manual review by an administrator is the deliberate trade the proposal makes for trustworthiness in a first version, and it is the single gate every other role's verified status passes through.
Live case map and rescue coordination	Views all active rescue cases plotted on a map, monitors their AI-assigned urgency and current status, and follows each one through to resolution.	Coordinating several simultaneous rescues is the operational task the whole rescue radar module exists to support, and the map is where that coordination actually happens.
AI triage oversight	Reviews and, where necessary, overrides Gemini's urgency classification for a reported case, alongside the same override available to a veterinarian.	This is the fallback the proposal names explicitly as the safeguard against an inaccurate AI classification turning into a wrong decision.
Content moderation	Reviews and removes adoption listings, messages, or profiles that violate platform rules or raise a fraud concern.	A trust network only holds together if bad-faith activity is removed promptly, which makes moderation part of maintaining the reputation model rather than a separate concern from it.
User account management	Activates, suspends, or removes an account, including one already verified, if evidence of misuse appears.	Verification is not a one-time guarantee of good behavior, so the administrator needs a way to act on misuse discovered after the fact.
Reputation and dispute handling	Adjusts reputation points or trust badges manually where a dispute between users requires it.	An automated reputation system still needs a human backstop for the cases it cannot resolve on its own, which is what keeps the reputation model fair rather than mechanical.
Admin dashboard and reporting	Sees a summary of pending verifications, open cases, and overall platform activity from a single web dashboard.	This dashboard is the operational home for every other administrator functionality on this list, and without it the individual functions above would have no shared starting point.



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Date of the status:

Approved by:

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Happy Paws

A mobile application for animal **rescue** and **rehoming**



A Project Proposed by

Higher Diploma in Computing and Software Engineering Students at
International College of Business and Technology (ICBT) Gampaha Branch
Batch: GM/HDCSE/CMU/18 (Group 01)



Acknowledgement

We thank **Ms. Erangi** for guiding us through our program and into the initial stages of our Computing Project. She provided constant support from our very first days to the submission of this proposal. Her clear feedback helped us understand what makes a project both ambitious and achievable.

She taught us exactly **how to structure a strong proposal and define the requirements for our future work**. We also learned how to build a realistic completion plan to keep our upcoming development on schedule. As we drafted this document, her advice on anticipating problems and finding practical solutions helped us plan with confidence.

We also thank our teammates for their dedication, adaptability, and hard work in preparing this document. Every member brought unique ideas to the table, shared the research workload, and kept the team motivated. This proposal reflects our shared commitment to building a successful final product in the coming months.

Finally, we appreciate the broader technical community for their invaluable open-source tools, tutorials, and documentation. Countless online forums, articles, and technical guides helped us shape our architecture and choose the right technologies. We rely on the collective knowledge shared by developers around the world to bring this proposal to life.



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1. Introduction

HappyPaws.lk is a proposal for a mobile platform that coordinates animal rescue and rehoming in Sri Lanka, but the claim behind it is narrower than that description suggests. The problem worth solving is not the absence of a way to report a stray, because local services such as SunaCare already offer exactly that (SunaCare, n.d.). The problem is the absence of trust. People who rescue, adopt, foster, and donate have no dependable way to know who is on the other end of a conversation, and that uncertainty is what this project sets out to remove. The argument of this proposal is that a verified-identity and reputation layer, paired with private communication, is the part of the system that justifies building it at all, and the scope follows from that position.

Two decisions shape everything that follows. The first is a deliberately small Minimum Viable Product (MVP) that a small team can build and show within twelve weeks, a constraint that rules out any feature the team cannot reuse or buy in as a managed service. The second is candor about overlapping with existing tools, because a proposal that claims to have no competitors invites the obvious rebuttal. The sections below set out the background, the problem, a review of comparable systems, the proposed solution, the reasoning behind it, its scope, the methodology and technology, and a twelve-week schedule with the risks that could derail it.

2. Background study

The animal-welfare community in Sri Lanka is active, generous, and poorly served by the tools it depends on. Rescue coordination, rehoming, and medical fundraising happen mostly through Facebook groups, WhatsApp threads, and general classifieds, none of which were designed for the job. On those channels an urgent case competes for attention with unrelated posts, and the people most able to help often never see it in time. The activity is not lacking in effort. It is lacking in infrastructure.

Some of that infrastructure has begun to appear, and any honest proposal has to say so. SunaCare lets a member of the public photograph a stray, captures the location automatically, alerts verified non-governmental organizations, and plots the case on a live map, all from a browser (SunaCare, n.d.). Bully Pully connects rescuers with prospective adopters in and around Colombo (Bully Pully, n.d.), and established charities such as Animal SOS Sri Lanka provide veterinary care and rehoming through their own sites (Animal SOS Sri Lanka, n.d.). These are real improvements on a plain social-media post, which means the reporting problem is already partly solved before this project starts.

The picture beyond Sri Lanka points the same way. Applications such as Pawrpose add urgency levels and live alerts to nearby volunteers (Pawrpose, n.d.), and recent academic work treats AI-assisted rescue reporting as a settled design rather than a research frontier, pairing image recognition with location-based triage to speed up response (ICAAAI, 2025, and IJCRT, 2025). The lesson is worth stating plainly. Photographing an injured animal and routing it to responders no longer sets a product apart, so a proposal that leads with that capability is proposing to rebuild what already exists.

What none of these systems provide is a shared sense of who their users are. Reporting, rehoming, fundraising, and shelter administration each sit in separate tools with separate accounts, so trust cannot accumulate anywhere. A fair objection is that combining these functions is only bundling, and bundling is not innovation. The answer is that the value does not come from placing features side by side. It comes from a single verified identity that carries a reputation across rescues, adoptions, and donations, which no individual tool can produce on its own. That shared identity is the gap this project targets.

3. Problem statement

The difficulties animal lovers face look like separate problems, but most of them trace back to one missing piece. Rehoming an animal or asking for help usually means posting a personal phone number in public, which exposes owners and rescuers to spam, scams, and unwanted contact. When an animal is injured, the appeal sinks into a social feed with no prioritization and no way to reach responders by location, so the time that matters most is lost. Neither problem is really about phones or feeds. Both follow from a system that has no idea who anyone is.

The same root shows up elsewhere. Without verification, a user cannot tell a committed adopter from an impulsive one, a reliable foster from someone an animal should never go to, or a genuine medical appeal from fraud. Advice about diet or behavior arrives from whoever comments first rather than from a veterinarian, and donations are sent to personal accounts with no evidence that the money reached the animal. Each of these is a failure of trust rather than a failure of features. That is the reason the proposed solution treats verified identity and reputation as the foundation, and everything else as something built on top of it.

3.1 Aim

To design and develop HappyPaws.lk, a verified-identity and reputation-led mobile platform that closes the specific trust gap in Sri Lankan animal rescue and rehoming, rather than resolving the reporting problem existing tools already handle. The system pairs AI-assisted photo triage and geo-targeted alerting with role-based identity verification, a reputation ledger, and private in-app messaging, so that adopters, fosters, transporters, sponsors, and veterinarians can coordinate a rescue, an adoption, or a donation around a shared, accountable identity instead of an anonymous post.



3.2 Objectives

- To identify why existing rescue, rehoming, and donation channels in Sri Lanka fail to give animal welfare volunteers a dependable way to know who they are dealing with.
- To review local and international rescue and rehoming platforms and establish, through direct comparison, where a verified-identity and reputation layer differs from what they already offer.
- To design and implement a role-based registration and identity verification system covering adopters and owners, fosters, transporters, sponsors, and veterinarians, with administrator review built into the workflow.
- To implement AI-assisted photo triage and geo-targeted alerting so a reported rescue case reaches verified, nearby responders without delay.
- To implement a secure adoption hub with lifestyle-based matching and private in-app messaging that removes the need for any user to publish a personal phone number.
- To evaluate, within the twelve-week MVP, whether the resulting platform lets adopters, fosters, transporters, sponsors, and veterinarians coordinate rescue, fostering, transport, and adoption activity around a verified identity.

4. Literature review

This review looks at the systems and research closest to HappyPaws.lk in order to place it accurately, rather than to flatter it. Read carefully, the existing work does not leave a wide-open field. It leaves a specific gap and naming that gap honestly is more useful than listing features no competitor has.

4.1 Local rescue and rehoming platforms

SunaCare is the fairest comparison for this project, and the most uncomfortable one. It already handles photo-based reporting with automatic location capture, alerts to verified organizations, and a public live map (SunaCare, n.d.), which covers much of what an animal-rescue app is usually asked to do. Where it stops is instructive. It does not give individuals verified roles, it does not let reputation accumulate, it does not carry private messaging between users, and it does not match adopters to animals. Bully Pully connects rescuers and adopters but relies on manual coordination and ordinary contact details (Bully Pully, n.d.), and the Facebook groups and classifieds most people still use trade privacy, verification, and any sense of priority for reach.

4.2 International and academic solutions

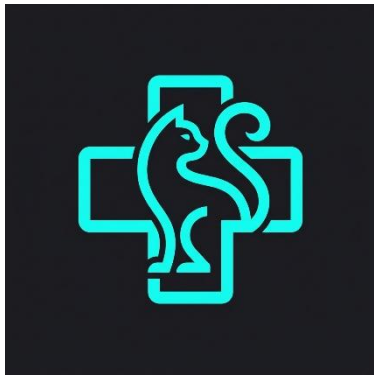
Looking further afield sharpens the point rather than softening it. Pawrpose layers urgency levels and volunteer alerts onto rescue reports (Pawrpose, n.d.), and research from 2025 treats an image-recognition model feeding a triage pipeline as a standard component rather than a novel one (ICAAAI, 2025, and IJCRT, 2025). Shelter-management products, for their part, concentrate on internal records rather than public coordination. Taken together, the literature shows that triage and reporting are mature, and that the unsolved work sits around them, in identity and trust, not in the camera.

4.3 The gap and this project's position

HappyPaws.lk is therefore positioned as a trust network for animal welfare rather than another reporting tool, and the comparison below is deliberately honest about where existing services already win. The point of the table is not to claim that nothing else works. It is to show that the one column HappyPaws.lk occupies alone is the identity-and-trust column, which is the column this proposal argues actually matters.

Capability	HappyPaws.lk (MVP)	SunaCare	Bully Pully	FB Classifieds /
Photo and location injury reporting	Yes	Yes	Partial	No
AI photo-based urgency triage	Yes	No	No	No
Geo-targeted alerts to verified responders	Yes	Yes (NGOs)	No	No
Verified roles and reputation	Yes	No	No	No
Private in-app chat, no phone sharing	Yes	No	Limited	No
Structured adoption lifestyle matching	Yes	No	Manual	No
Transparent direct-to-vet funding	Roadmap	No	No	No
Community and expert Q and A	Roadmap	No	No	Unstructured

5. Proposed solution



HappyPaws.lk will be a native Android application for the public, with a web administration panel for organizations, verified veterinarians, and administrators. Given a twelve-week build, the design has to be strict about what counts as core, so the MVP is limited to three modules that together form one working path from a verified account to a coordinated rescue. Anything that does not serve that path waits for a later phase. The three modules were not chosen because they look impressive. They were chosen because each one carries part of the trust argument the rest of this proposal makes.

5.1 Verified roles and trust

Every user registers under a role, as an adopter or owner, a foster, a transporter, a sponsor, or a veterinarian, and confirms identity by submitting a document that an administrator reviews through the web panel. Reputation then accumulates from confirmed activity, such as completed adoptions and verified rescues, and shows up as trust badges. This is the module that makes the platform more than a listing board, because it is the only place where a person's history follows them from one interaction to the next.

5.2 Secure adoption hub and private messaging

Adopters complete a short lifestyle profile covering home size, activity level, and existing pets, and a rule-based engine suggests animals that fit. All contact runs through in-app messaging built on SignalR, so no one has to publish a phone number to arrange a meeting. Rule-based matching is a considered choice over machine learning here, because twelve weeks does not allow for collecting and labelling the data a model would need, and a transparent set of rules is easier to explain to an examiner and a user than a trained model would be.

5.3 Rescue radar with AI photo triage

When a user finds an injured animal, they photograph it in the app while the device records the location, and Google's Gemini vision API classifies the case as critical, moderate, or low urgency. Verified volunteers, fosters, and transporters near the incident receive a push notification through Firebase Cloud Messaging, and the case appears on the administration map. Using a hosted model rather than a custom one is what makes a real AI feature possible inside the schedule, and treating its output as advice rather than instruction keeps a wrong classification from turning into a wrong decision.

5.4 MVP scope with MoSCoW prioritization

A twelve-week window makes prioritization the most important design activity, not an afterthought, so the MVP is scoped with MoSCoW and the boundaries are drawn tightly. The Must-have row is the whole product. The rows beneath it are what the team will add only if the core is finished with time to spare.

Priority	Features
Must have	Registration with a role, administrator identity verification, rescue reporting with photo and location, AI photo triage, geo-targeted alerts to verified responders, private in-app messaging, adoption listings with a lifestyle profile and rule-based matching, reputation and trust badges, and a web administration panel for verification and case management.
Should have	Push notifications for alerts and messages, in-app case status tracking, and search and filtering of listings and cases.
Could have	Gamified leaderboards, multi-language support for Sinhala, Tamil, and English, and a dark theme.
Won't have (this cycle)	Real payments and crowdfunding, a machine-learning matchmaking model, AI facial-recognition lost-and-found, a public question-and-answer community, and an iOS application.

5.5 Post-MVP roadmap

The features left out are sequenced rather than abandoned, and the order reflects risk as much as value. A second phase adds direct-to-vet crowdfunding with itemized, transparent fund tracking, together with a knowledge hub where verified veterinarians answer questions and earn expert standing. A third phase takes on the work that genuinely needs data and time, namely a machine-learning matchmaking model trained on the platform's own outcomes, an AI facial-recognition service for lost and found animals, and an iOS build. Holding real payments and custom models back is what lets the first twelve weeks stay honest about what can be finished.



6. Justification of the solution

The case for building around trust rather than reporting rests on the competitive picture already described. If reporting and triage are handled well by others, then spending a short project rebuilding them produces something with no reason to be chosen over SunaCare. Concentrating on verified identity, private communication, and reputation puts the effort where no local service currently competes, and where the risks users worry about, namely fraud, abuse, and exposure of personal data, are met directly.

Choosing native Android over a cross-platform toolkit deserves a defense, because cross-platform frameworks usually promise to save time. The rescue path depends on the camera, location, and reliable push notifications, and native Android gives the most direct and dependable access to all three. That reliability matters more than a few saved days when the feature has to work in front of an examiner. A React and TypeScript admin panel then gives organizations and veterinarians a proper desktop surface for verification and case management, work that does not belong on a phone.

On the server, an ASP.NET Core Web API with PostgreSQL, hosted on Amazon Web Services, was chosen for reasons of both capability and time. PostgreSQL with the PostGIS extension answers proximity queries efficiently, which the alerting feature needs, and managed AWS services for compute, database, and storage remove setup work that a twelve-week schedule cannot absorb. Gemini handles triage for the same reason, rule-based logic handles adoption. Buying a capable hosted model is faster and steadier than training one, and the goal here is a working feature rather than a research contribution.

The strongest argument for the plan is the scope itself. A twelve-week MVP governed by MoSCoW, with payments and custom machine learning deferred, is not a shrunken version of an ambitious project. It is a bet that a narrow, working, trustworthy slice will convince a reader more than a broad prototype that does nothing from end to end. That bet is the real proposal, and the rest of the design exists to protect it.



7. Scope of the system

7.1 In scope for the MVP

The MVP covers one continuous path and the pieces that path needs. A user can register under a role and be verified by an administrator, report an injured animal with a photo and location, and have that report triaged and pushed to nearby verified responders. The same account can list an animal for adoption, complete a lifestyle profile, receive rule-based suggestions, and talk to another user privately in the app. Administrators get a web panel for verification, moderation, and live case management, which is the minimum the rest of the system needs to run safely.

7.2 Out of scope, deferred to the roadmap

Several visible features are intentionally out of scope for the twelve weeks. There is no real payment processing or crowdfunding, no machine-learning matchmaking, no facial-recognition lost-and-found, no public question-and-answer community, and no iOS build. These are not rejected ideas. They are the roadmap, held back so that the core can be finished rather than left half-built across a wider surface.

7.3 System users

The platform serves adopters and owners, fosters, transporters, sponsors and donors, veterinarians, and the administrators who run verification. Each role sees only the functions and data that suit its responsibilities and its verification status, which is part of how the trust model keeps its meaning.

7.4 Assumptions and constraints

The plan assumes that users have an Android device with a working camera, location, and connection, and that the Gemini API stays available within its quota. It assumes that administrators can review verification requests quickly enough to keep the queue short, since manual review is the deliberate trade made for trustworthiness in a first version. It also assumes that no real money moves through the MVP, so any funding shown in a demonstration is simulated. The constraint behind all of these is time, and the scope has been drawn to respect it.

8. Methodology and technology

8.1 Development methodology

The project uses Agile delivery, adapted to the timeline as six two-week sprints, each ending in something that runs. This suits a project whose details will shift as the team learns, and it pairs naturally with the MoSCoW ordering, because each sprint pulls the next most important item off the list. Design is not skipped, but it is kept light and mostly done up front, using object-oriented analysis and design with a small set of UML use-case, class, and sequence diagrams to hold the structure together while the code moves quickly.

8.2 System architecture

HappyPaws.lk uses three-tier client-server architecture. The Android application and the React admin panel form the presentation tier and talk over HTTPS to an ASP.NET Core Web API, which holds the business logic, authentication, and real-time messaging. A PostgreSQL database with PostGIS is the data tier. The external services the system depends on, Gemini for triage, Firebase Cloud Messaging for alerts, and Amazon S3 for media, are reached through the API so that credentials and coordination stay on the server rather than on the device.

8.3 Technology stack

The stack below reflects the two priorities that run through this proposal, dependable access to device features and as little hand-built infrastructure as the schedule allows.

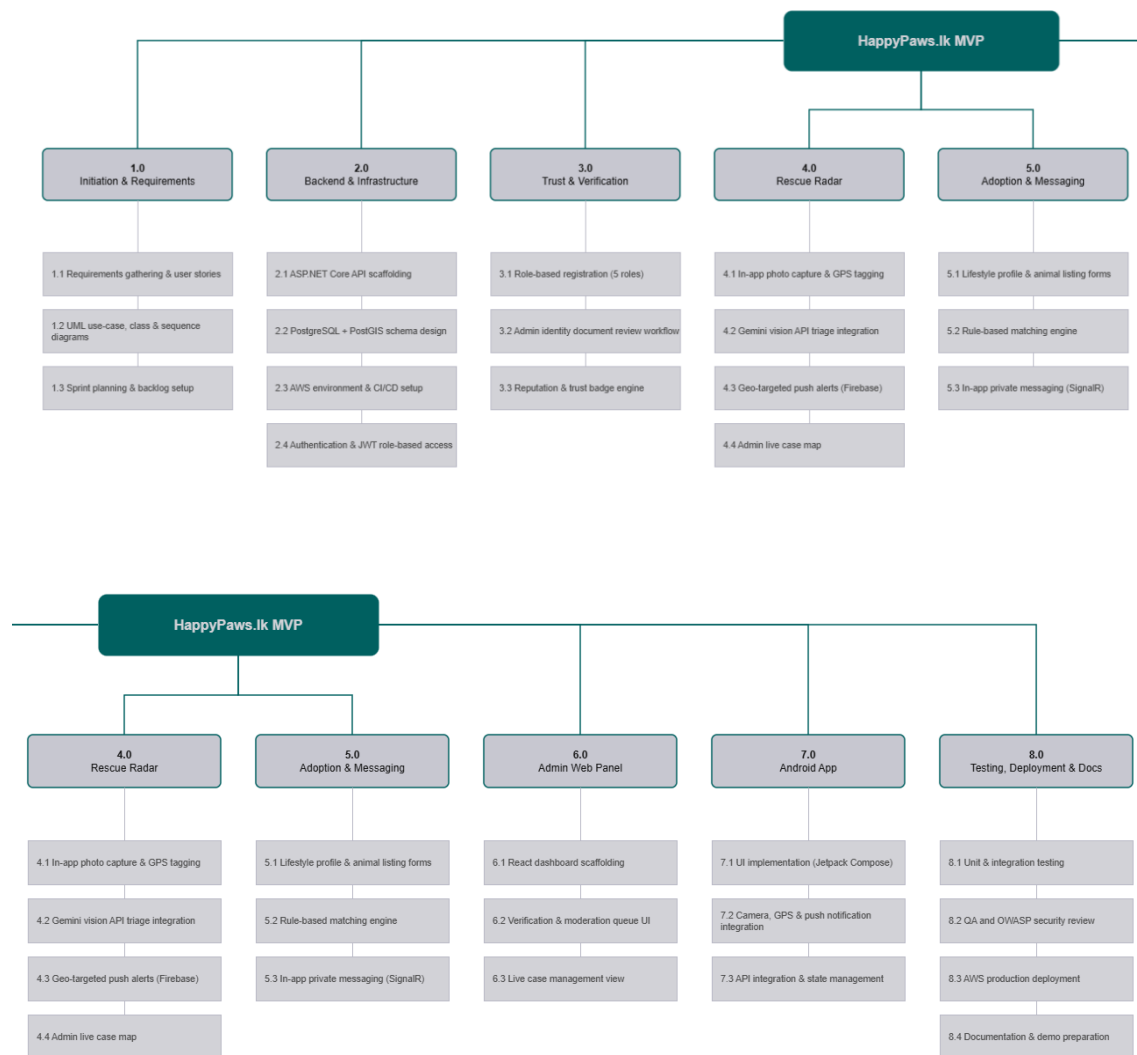
Layer	Technology	Purpose
Mobile application	Kotlin, Jetpack Compose, Google Maps SDK, CameraX	Native Android user app for reporting, maps, camera, and messaging.
Administration panel	React, TypeScript	Web dashboard for verification, moderation, and case management.
Backend and API	ASP.NET Core Web API (C#), Entity Framework Core, SignalR	Business logic, REST endpoints, and real-time in-app chat.
Database	PostgreSQL with PostGIS	Persistent storage and geospatial proximity queries.
AI service	Google Gemini vision API	Photo-based urgency triage of rescue reports.
Authentication	ASP.NET Identity with JWT	Secure, role-based access control.

Layer	Technology	Purpose
Notifications	Firebase Cloud Messaging	Geo-targeted rescue and message alerts.
Deployment and infrastructure	AWS (EC2 or Elastic Beanstalk, Amazon RDS, Amazon S3, CloudWatch)	Hosting for the API and database, media storage, and monitoring.
Tooling	Git and GitHub, Android Studio, Visual Studio, Postman, Figma, Jira	Version control, development, API testing, design, and tracking.

9. Gantt chart and risks

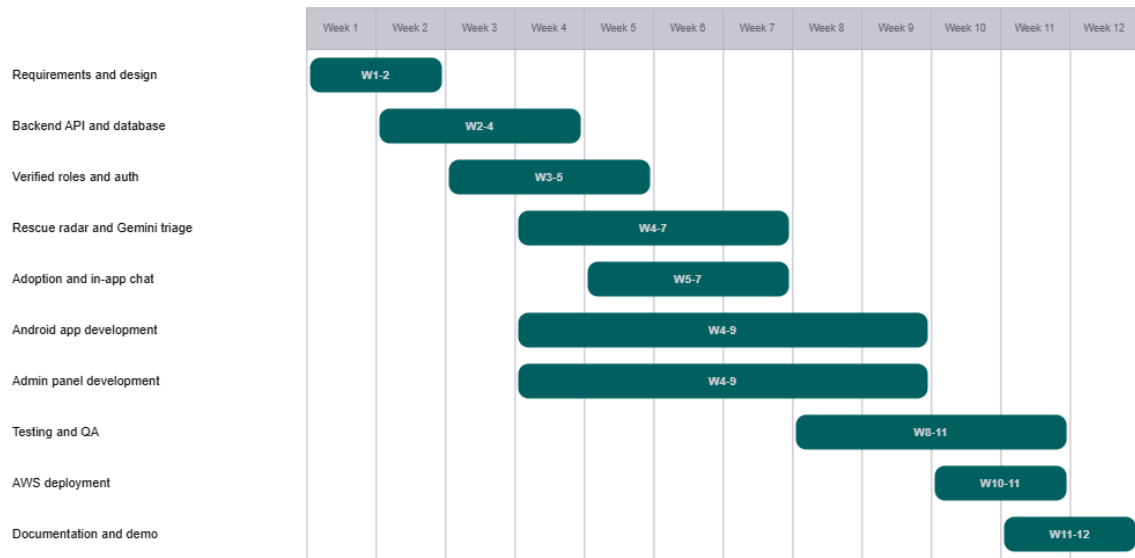
9.1 Work breakdown structure

The MVP breaks down into eight work packages and twenty seven tasks, grouped by the module each task serves rather than by role, so the dependency between a task and the trust argument it supports stays visible.



9.2 Project schedule

The schedule fits the MVP into twelve weeks, run as six two-week sprints, with the phases below overlapping once the initial design work is done. Building in parallel is what makes the twelve weeks enough to deliver every must-have functionality described in section 8, not just a subset of it. The Gantt chart below will be used to track the schedule during delivery.



Bars show planned active weeks per phase. Overlapping bars indicate parallel workstreams once the design phase (Weeks 1-2) is complete.

9.3 Risk analysis

The risks below are ordered by how much they threaten a finished demonstration, and the first of them is the twelve-week schedule itself. Naming it as the dominant risk is more useful than pretending the timeline is comfortable, because every mitigation that follows is really a way of protecting the same deadline.

Risk	Likelihood	Impact	Mitigation
The twelve-week timeline itself	Medium	High	Keep to the Must-have items, reuse managed services and libraries, and cut Could-have features first if a sprint slips.
Scope creep beyond the MVP	Medium	High	Hold the MoSCoW line and route every new idea to the roadmap rather than into the current build.
Gemini API cost, quota, or availability	Medium	Medium	Cache results, cap request rates, and fall back to a rule-based urgency estimate if the service is unavailable.
Inaccurate AI triage	Medium	Medium	Treat the classification as advice, allow administrator override, and check it against sample images.
Verification backlog	Medium	Medium	Prioritize the queue, allow several administrators, and add lightweight automatic checks.
Data privacy or security breach	Low	High	Encrypt data in transit and at rest, enforce role-based access with JWT, and follow OWASP guidance.
Team unfamiliarity with .NET or AWS	Medium	Medium	Spend part of the first sprint on setup and learning and prefer managed services to hand-built infrastructure.

10. References

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